

Visualization Draft

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1. Set up

Packages

Plot settings

Data

2. Clean data

Cleaning weather data

Goal

Clean weather dataset with date, precipitation, maximum temperature, minimum temperature, month, year, and water year.

Cleaning metadata

Goal

Clean metadata dataset with survey date, site information, water year, and readable site names.

Cleaning water parameter data

Goal

Clean water quality parameter dataset with elevation code, dissolved oxygen, temperature, salinity, and conductivity.

3. Join data frames

Water quality = metadata + water parameters

Water quality weather = water quality + weather

4. Filter data for dissolved oxygen analysis

Goal

Create a filtered dissolved oxygen dataset that focuses on the three main NCOS sites.

5. Summarize dissolved oxygen

Dissolved oxygen by site

```
# A tibble: 3 x 6
  site_full      mean_do median_do      sd_do var_do      n
  <chr>          <dbl>    <dbl>    <dbl>  <dbl> <int>
1 East Channel  304885.    8.10  5521576.  3.05e13  328
2 Phelps Bridge    5.62    5.40    3.32  1.10e 1  382
3 Venoco Bridge 1426725.    6.98 36189088.  1.31e15  771
```